



Faculty of Computer Applications

**BCA Sem-1
(AY-2026-27)**

Lab Book

**Sub Code : 05BC3103
(Programming Practices : 1)**

Roll No :	92600527015
Full Name :	Harshrajsinh Jadeja

GitHub Link | [harshrajjadeja701-coder \(Harshrajsinh-701\)](https://github.com/harshrajjadeja701-coder)

Sub.Faculty Name : Dr.Ashwin Dobariya

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4.	Write a program that input value from user and find out Area of Circle.
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9.	Write a program Interchange the value of two numbers without using third variable.
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(1) Write a Program to Print "Hello World".

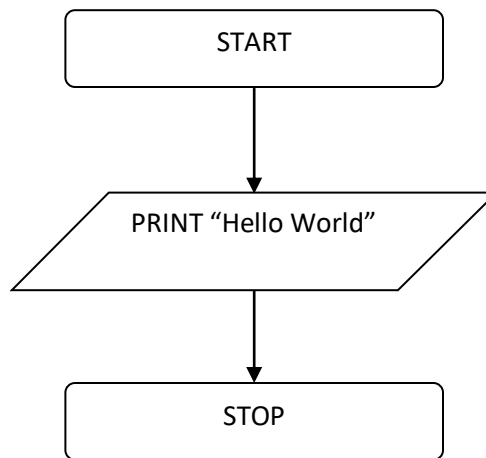
Algorithm :

Step 1 : START

Step 2 : PRINT "Hello World"

Step 3 : STOP

Flowchart :

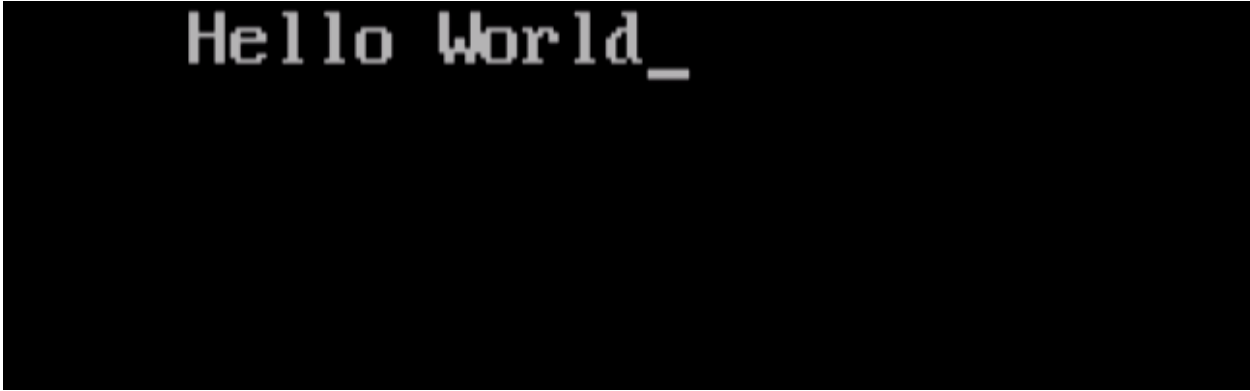


Code :

```
#include<stdio.h>
#include<conio.h>

void main(){
    clrscr();
    printf("Hello World");
    getch();
}
```

Output :

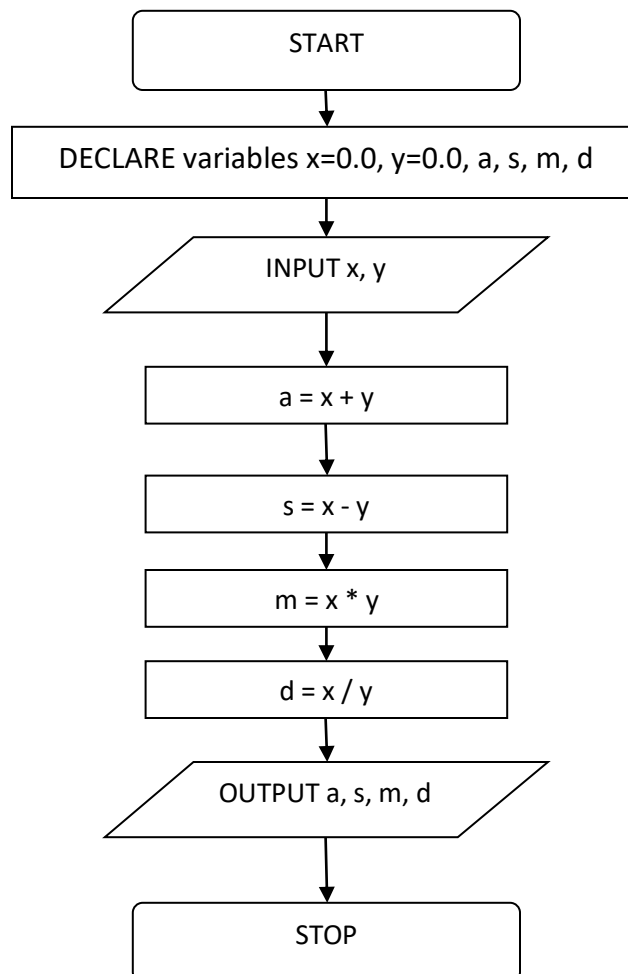
A screenshot of a terminal window with a black background. The text "Hello World_" is displayed in a light gray, monospaced font. The text is positioned in the upper left area of the terminal, followed by a cursor underscore.

(2) Write a Program that takes 2 Values from user and Perform All Arithmetic Operation.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables $x=0.0$, $y=0.0$, a , s , m , d
Step 3 : INPUT x, y
Step 4 : $a = x + y$
Step 5 : $s = x - y$
Step 6 : $m = x * y$
Step 7 : $d = x / y$
Step 8 : OUTPUT a, s, m, d
Step 9 : STOP

Flowchart :



Code :

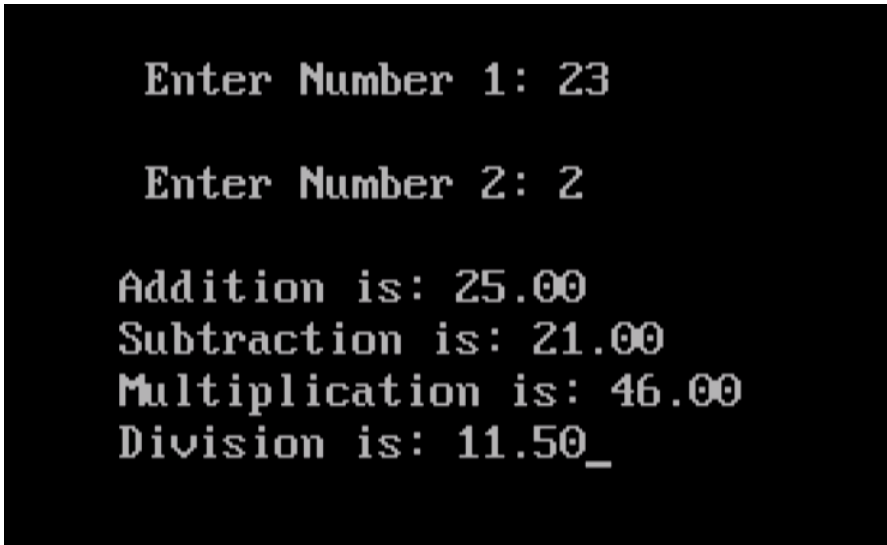
```
#include<stdio.h>
#include<conio.h>

void main()
{
    float x,y;
    clrscr();
    printf("\n Enter Number 1: ");
    scanf("%f",&x);
    printf("\n Enter Number 2: ");
    scanf("%f",&y);

    printf("Addition Is: %.2f \n",x+y);
    printf("Subtraction Is: %.2f \n",x-y);
    printf("Multiplication Is: %.2f \n",x*y);
    printf("Division Is: %.2f \n",x/y);

    getch();
}
```

Output :

A screenshot of a terminal window with a black background and white text. The output shows the results of a C program where two numbers, 23 and 2, are entered. The program then displays the addition, subtraction, multiplication, and division of these two numbers, formatted to two decimal places. The division result is followed by a cursor underscore.

```
Enter Number 1: 23

Enter Number 2: 2

Addition is: 25.00
Subtraction is: 21.00
Multiplication is: 46.00
Division is: 11.50_
```

(3) Write a program that takes 3 values for Principle Amount, Rate of Interest and No. of Years and find out Simple Interest.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables SI, P, R, T

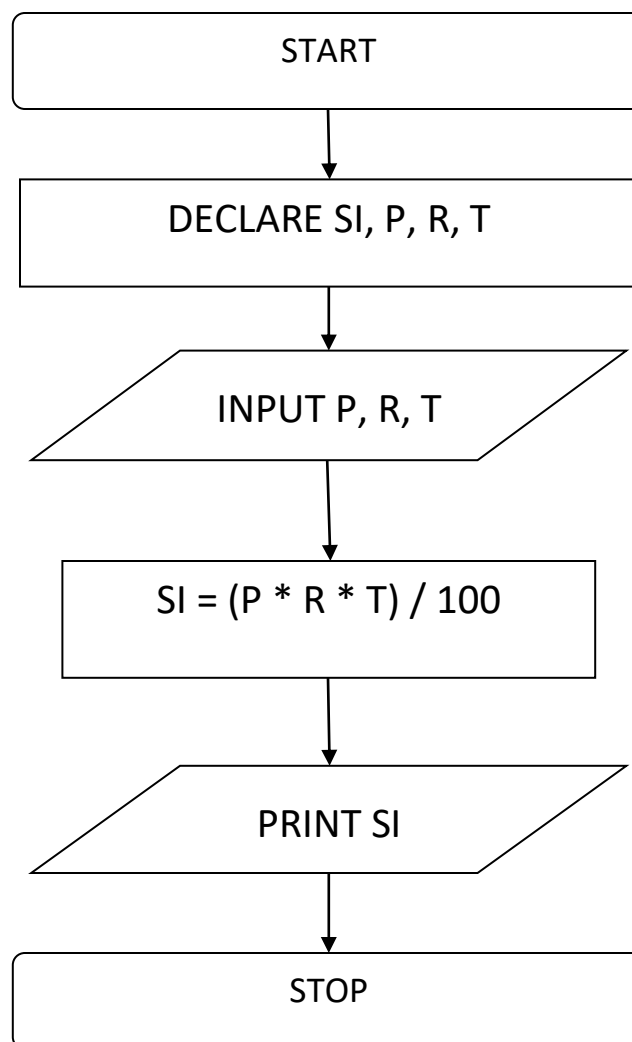
Step 3 : INPUT P, R, T

Step 4 : $SI = (P * R * T) / 100$

Step 5 : PRINT SI

Step 6 : STOP

Flowchart :



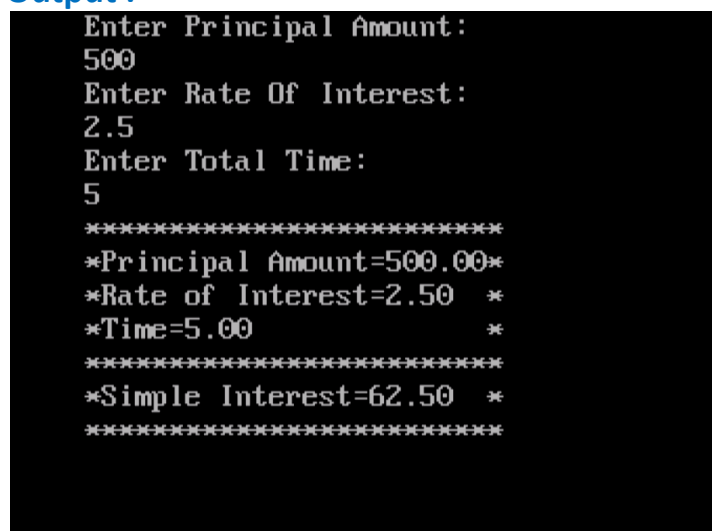
Code :

```
#include<stdio.h>
#include<conio.h>

void main(){
    float SI,P,R,T;
    clrscr();
    printf("Enter Principal Amount: \n");
    scanf("%f",&P);
    printf("Enter Rate Of Interest: \n");
    scanf("%f",&R);
    printf("Enter Total Time: \n");
    scanf("%f",&T);
    SI=(P*R*T)/100;
    printf("*****\n");
    printf("*Principal Amount=%.2f*\n",P);
    printf("*Rate of Interest=%.2f *\n",R);
    printf("*Time=%.2f      *\n",T);
    printf("*****\n");
    printf("*Simple Interest=%.2f *\n",SI);
    printf("*****");

    getch();
}
```

Output :



```
Enter Principal Amount:
500
Enter Rate Of Interest:
2.5
Enter Total Time:
5
*****
*Principal Amount=500.00*
*Rate of Interest=2.50 *
*Time=5.00      *
*****
*Simple Interest=62.50 *
*****
```

(4) Write a program that input value from user and find out Area of Circle.

Algorithm :

Step 1 : START

Step 2 : DECLARE variable r, area

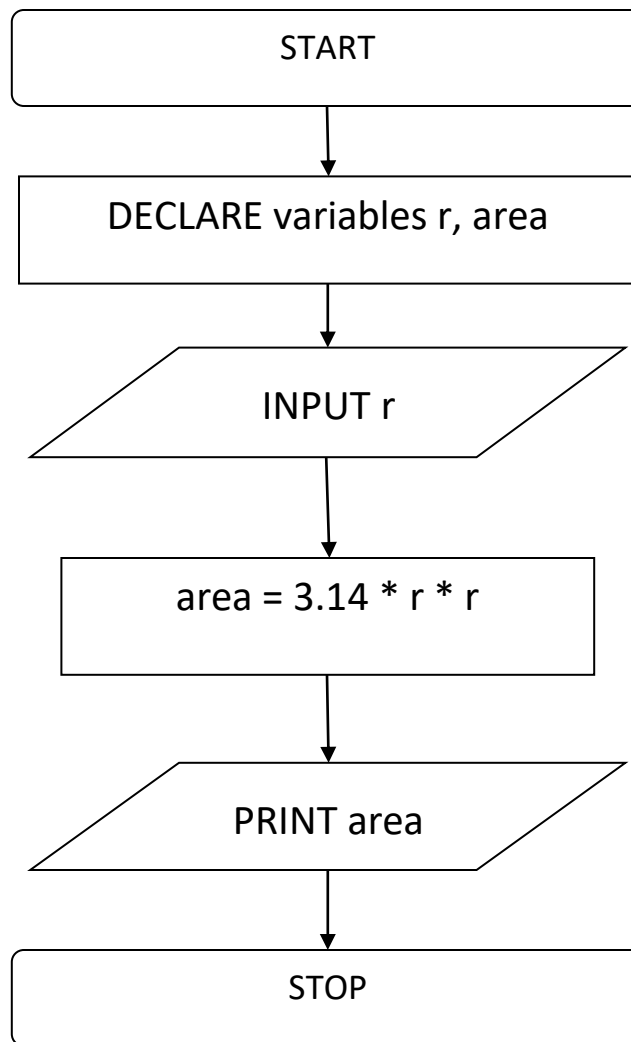
Step 3 : INPUT r

Step 4 : $\text{area} = 3.14 * r * r$

Step 5 : PRINT area

Step 6 : STOP

Flowchart :

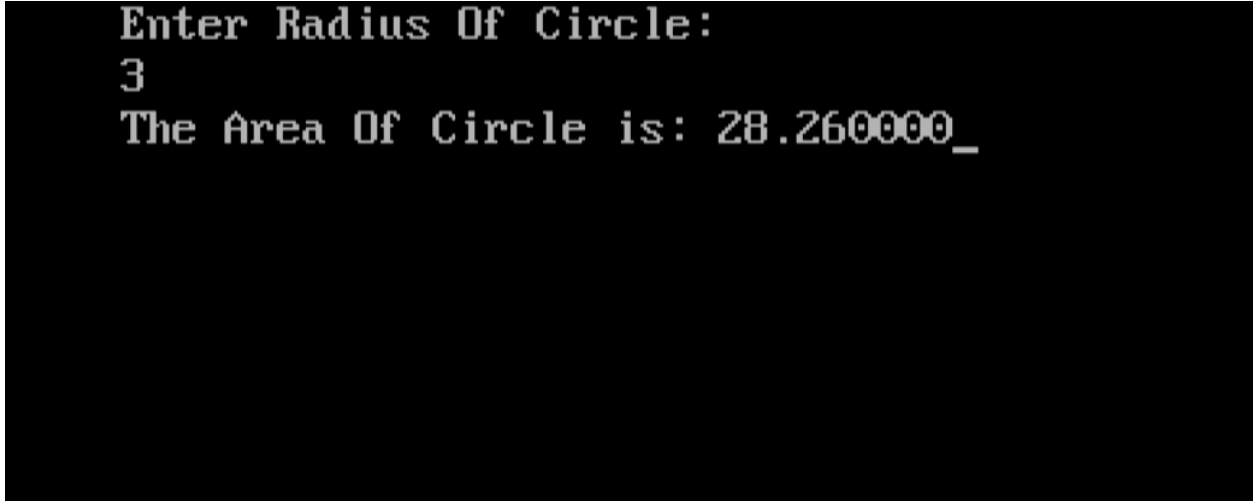


Code :

```
#include<stdio.h>
#include<conio.h>

void main()
{
    float Radius;
    clrscr();
    printf("Enter Radius Of Circle: \n");
    scanf("%f",&Radius);
    printf("The Area Of Circle is: %f",3.14*Radius*Radius);
    getch();
}
```

Output :



```
Enter Radius Of Circle:
3
The Area Of Circle is: 28.260000_
```

(5) Write a program that input two values from user and find out Area of Rectangle.

Algorithm :

Step 1 : START

Step 2 : DECLARE variable l, b, area

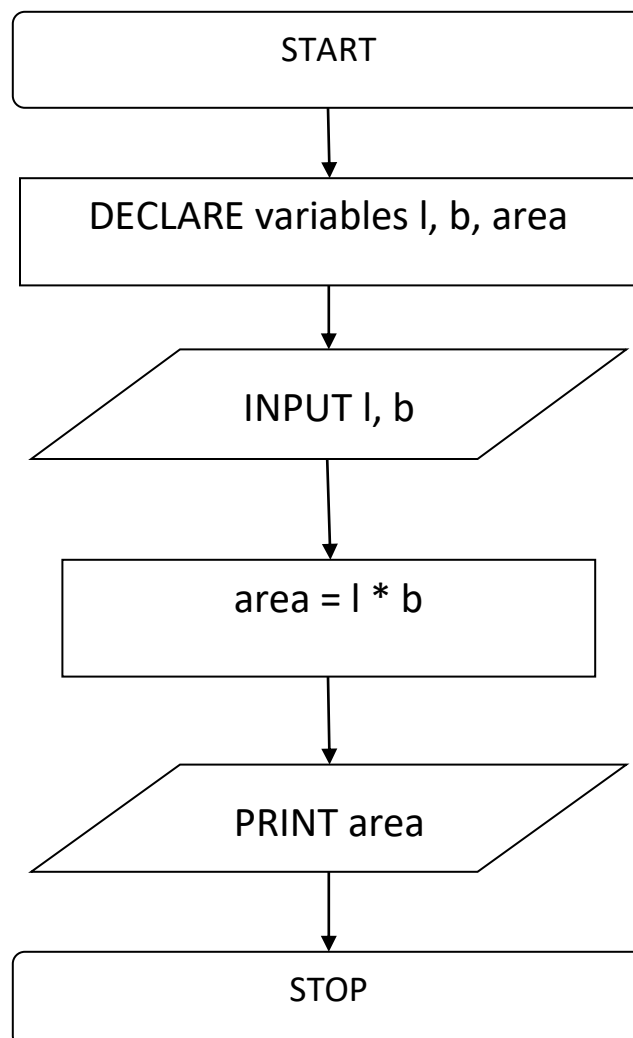
Step 3 : INPUT l, b

Step 4 : $\text{area} = l * b$

Step 5 : PRINT area

Step 6 : STOP

Flowchart :

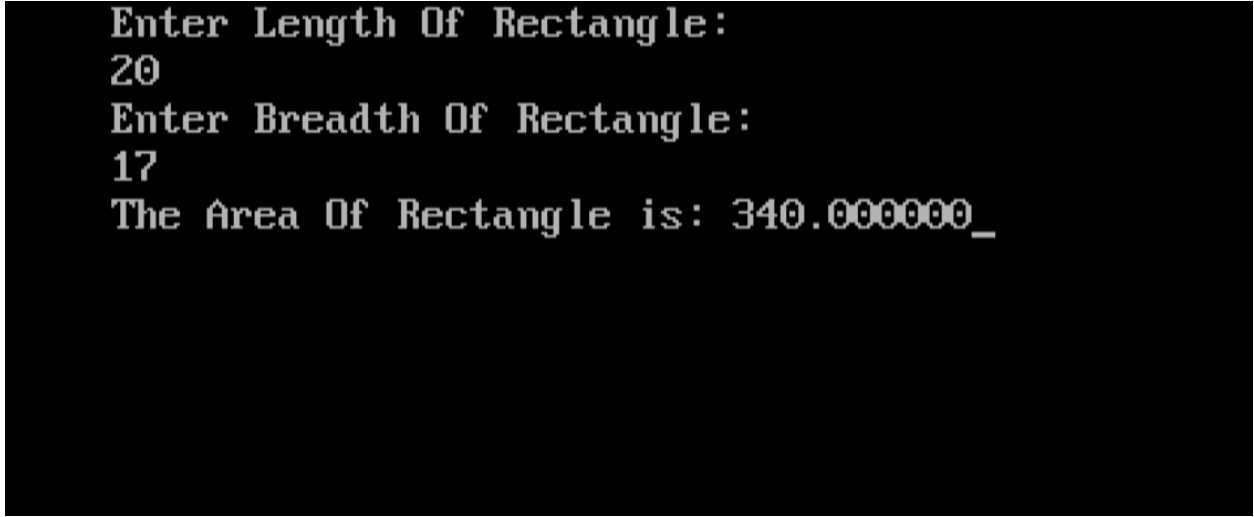


Code :

```
#include<stdio.h>
#include<conio.h>

void main()
{
    float Length,Breadth;
    clrscr();
    printf("Enter Length Of Rectangle: \n");
    scanf("%f",&Length);
    printf("Enter Breadth Of Rectangle: \n");
    scanf("%f",&Breadth);
    printf("The Area Of Rectangle is: %f",Length*Breadth);
    getch();
}
```

Output :



```
Enter Length Of Rectangle:
20
Enter Breadth Of Rectangle:
17
The Area Of Rectangle is: 340.000000_
```

(6) Write a program that input 3 values from user and find out its Average.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables x, y, z, avg

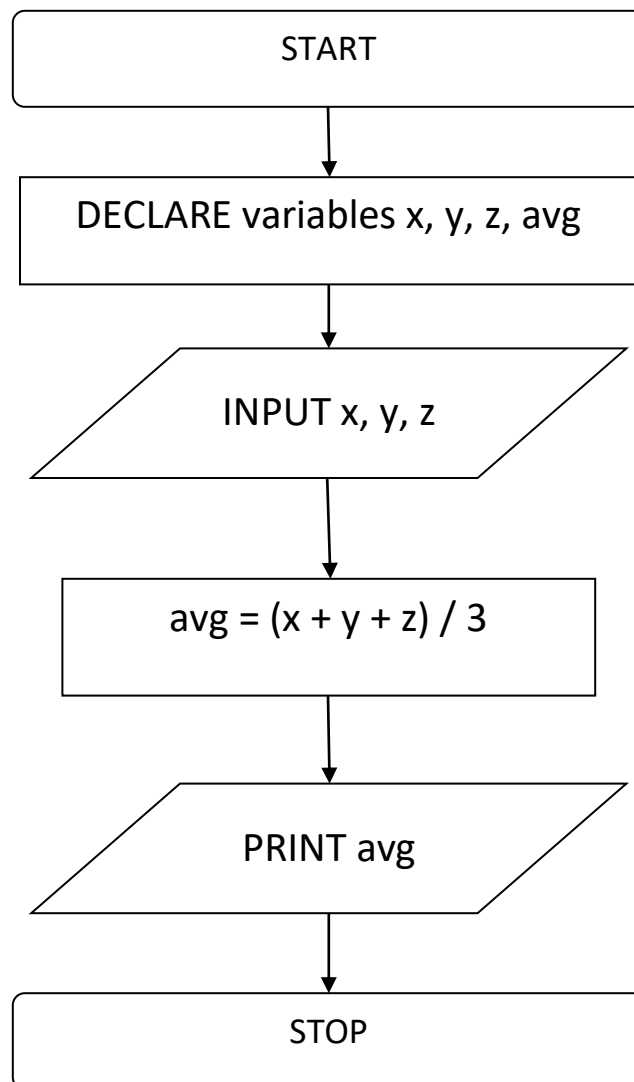
Step 3 : INPUT x, y, z

Step 4 : $\text{avg} = (x + y + z) / 3$

Step 5 : PRINT avg

Step 6 : STOP

Flowchart :

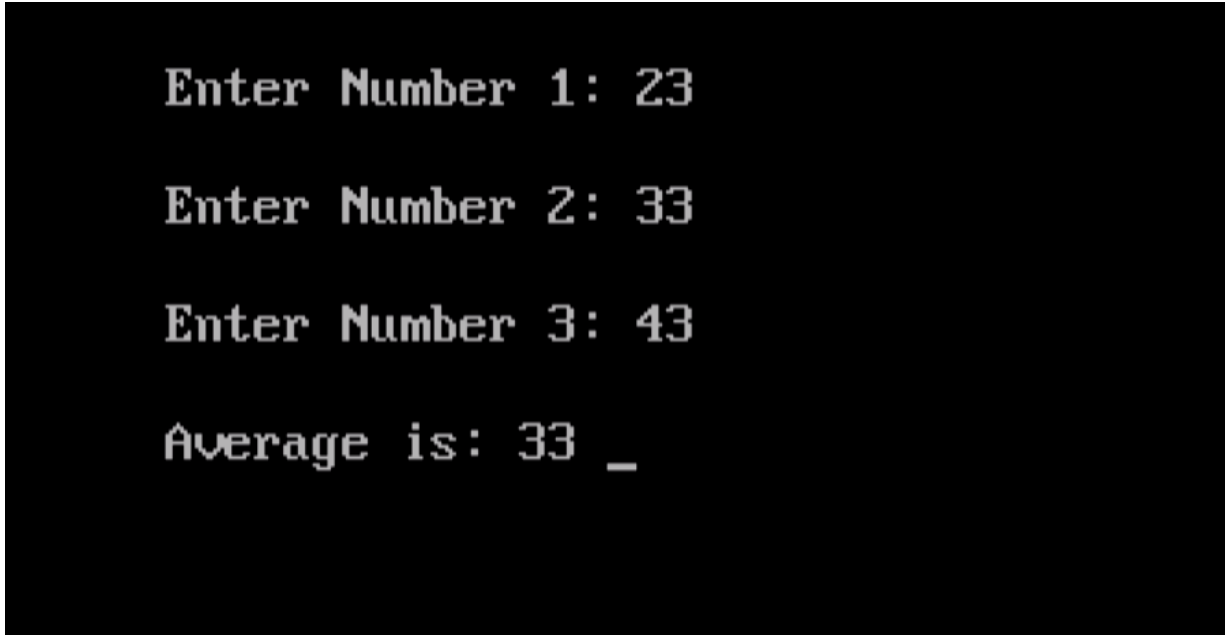


Code :

```
#include<stdio.h>
#include<conio.h>

void main()
{
    int x,y,z;
    clrscr();
    printf("\n Enter Number 1: ");
    scanf("%d",&x);
    printf("\n Enter Number 2: ");
    scanf("%d",&y);
    printf("\n Enter Number 3: ");
    scanf("%d",&z);
    printf("\n Average is: %d ",(x+y+z)/3);
    getch();
}
```

Output :



```
Enter Number 1: 23

Enter Number 2: 33

Enter Number 3: 43

Average is: 33 _
```

(7) Write a program to Calculate the Square and Cube of a number.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables n , sqr , $cube$

Step 3 : INPUT n

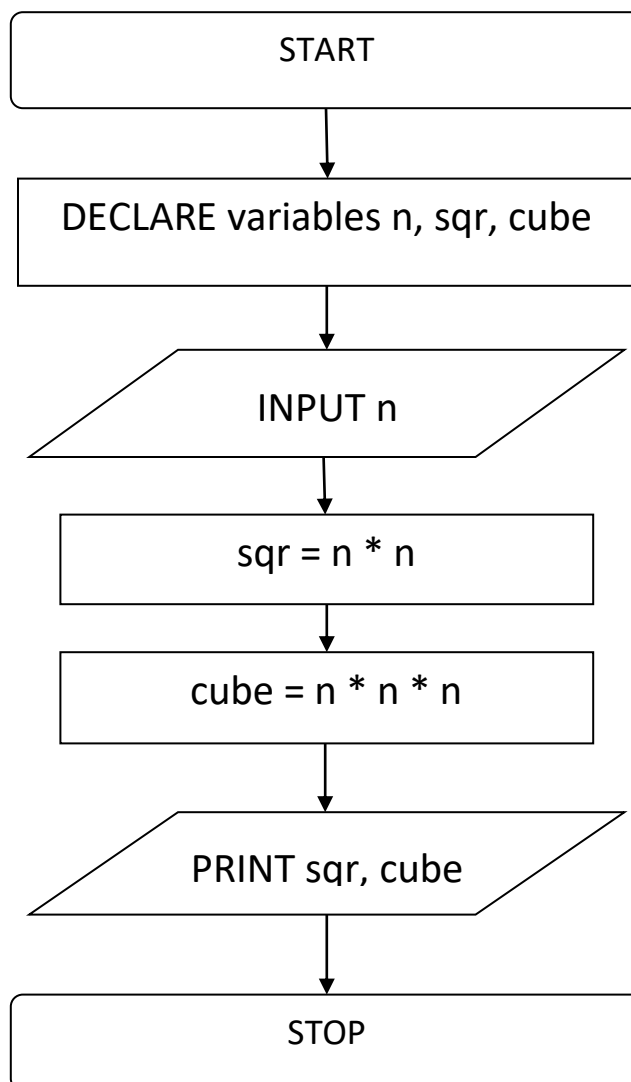
Step 4 : $sqr = n * n$

Step 5 : $cube = n * n * n$

Step 6 : OUTPUT sqr , $cube$

Step 7 : STOP

Flowchart :

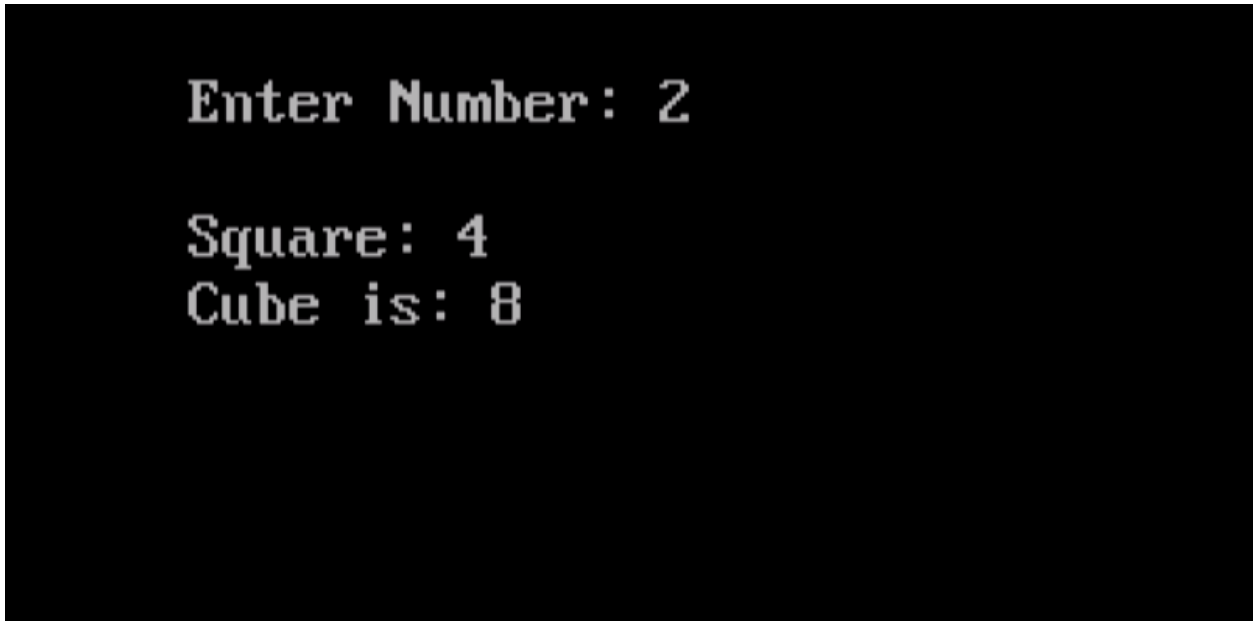


Code :

```
#include<stdio.h>
#include<conio.h>

void main()
{
    int n;
    clrscr();
    printf("\n Enter Number: ");
    scanf("%d",&n);
    printf("\n Square is: %d",n*n);
    printf("\n Cube is: %d",n*n*n);
    getch();
}
```

Output :



```
Enter Number: 2
```

```
Square: 4
```

```
Cube is: 8
```

(8) Write a program Interchange the value of two numbers using third variable.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables temp, x, y

Step 3 : INPUT x, y

Step 4 : temp = x

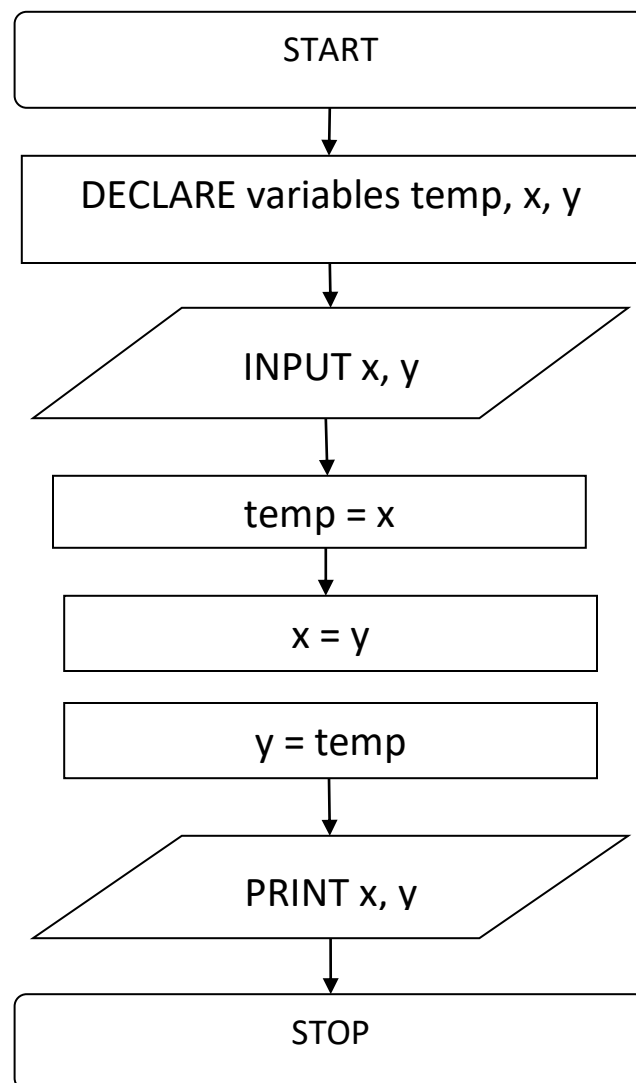
Step 5 : x = y

Step 6 : y = temp

Step 7 : PRINT x, y

Step 8 : STOP

Flowchart :

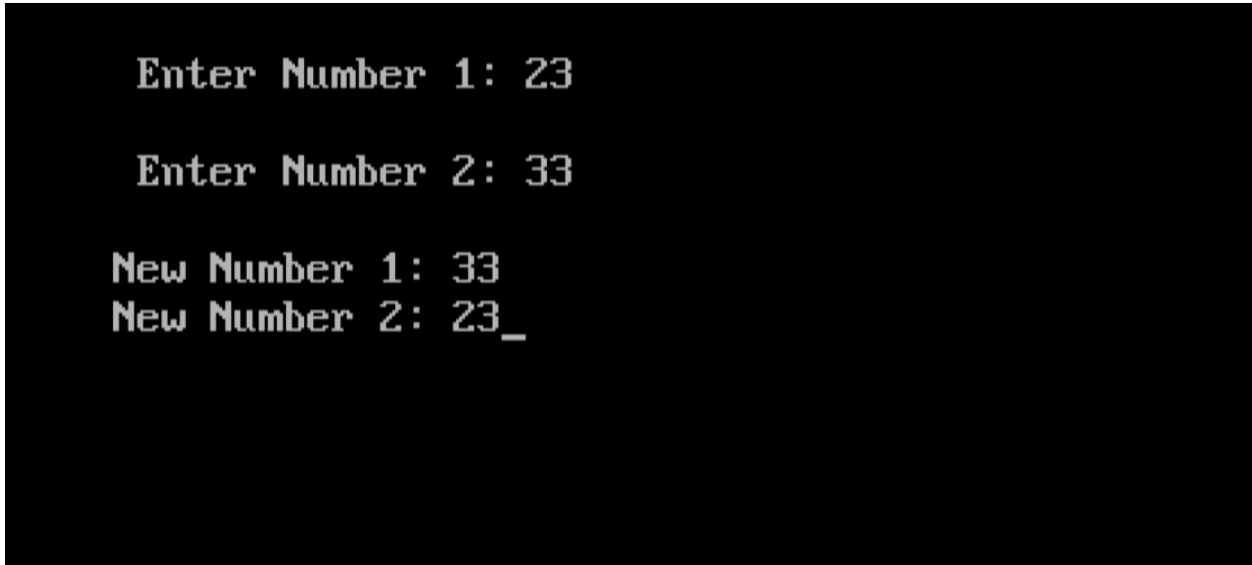


Code :

```
#include<stdio.h>
#include<conio.h>

void main()
{
    int temp,x,y;
    clrscr();
    printf("\n Enter Number 1: ");
    scanf("%d",&x);
    printf("\n Enter Number 2: ");
    scanf("%d",&y);
    temp=x;
    x=y;
    y=temp;
    printf("\nNew Number 1: %d",x);
    printf("\nNew Number 2: %d",y);
    getch();
}
```

Output :

A screenshot of a terminal window with a black background and white text. The output shows the program's execution: it prompts for 'Enter Number 1:' and receives '23', then prompts for 'Enter Number 2:' and receives '33'. It then displays 'New Number 1: 33' and 'New Number 2: 23_'.

```
Enter Number 1: 23

Enter Number 2: 33

New Number 1: 33
New Number 2: 23_
```

(9) Write a program Interchange the value of two numbers without using third variable.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables x, y

Step 3 : INPUT x, y

Step 4 : $x = x + y$

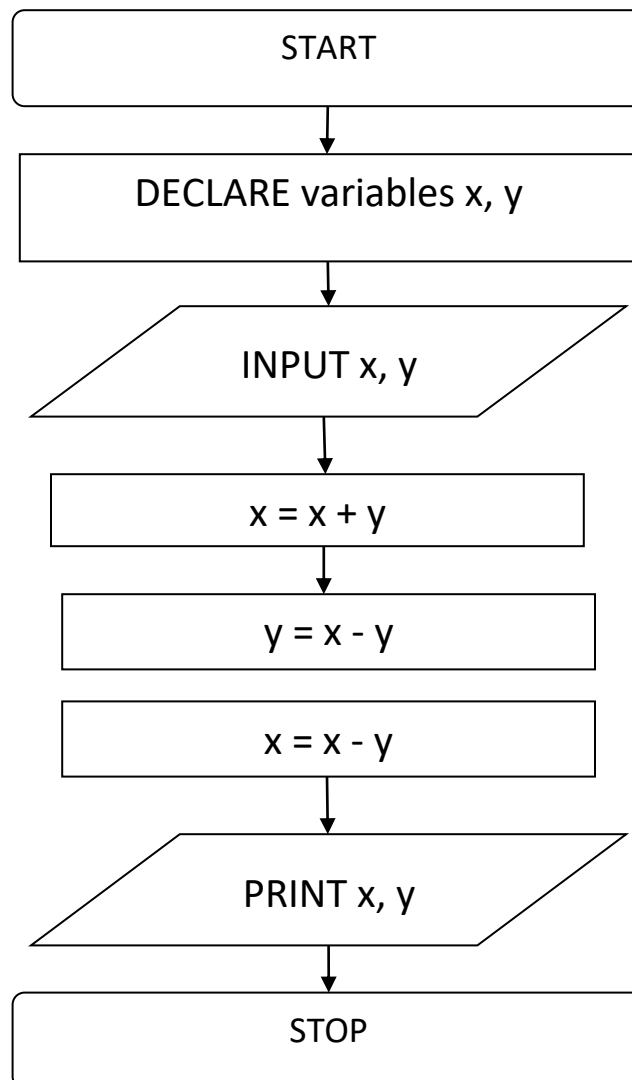
Step 5 : $y = x - y$

Step 6 : $x = x - y$

Step 7 : PRINT x, y

Step 8 : STOP

Flowchart :

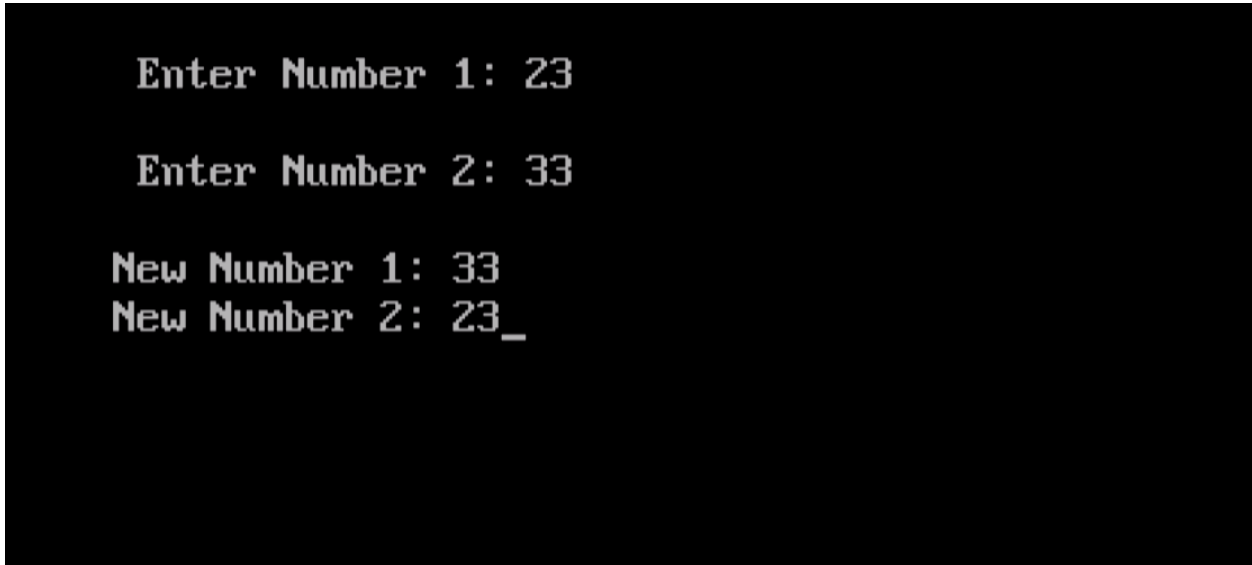


Code :

```
#include<stdio.h>
#include<conio.h>

void main()
{
    int x,y;
    clrscr();
    printf("\n Enter Number 1: ");
    scanf("%d",&x);
    printf("\n Enter Number 2: ");
    scanf("%d",&y);
    x=x+y;
    y=x-y;
    x=x-y;
    printf("\nNew Number 1: %d",x);
    printf("\nNew Number 2: %d",y);
    getch();
}
```

Output :

A screenshot of a terminal window with a black background and white text. It shows the output of the provided C program. The text is as follows:

```
Enter Number 1: 23

Enter Number 2: 33

New Number 1: 33
New Number 2: 23_
```

(10) Write a program that input quantity, price, discount and display final amount.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables q, pr, disc, f_amt

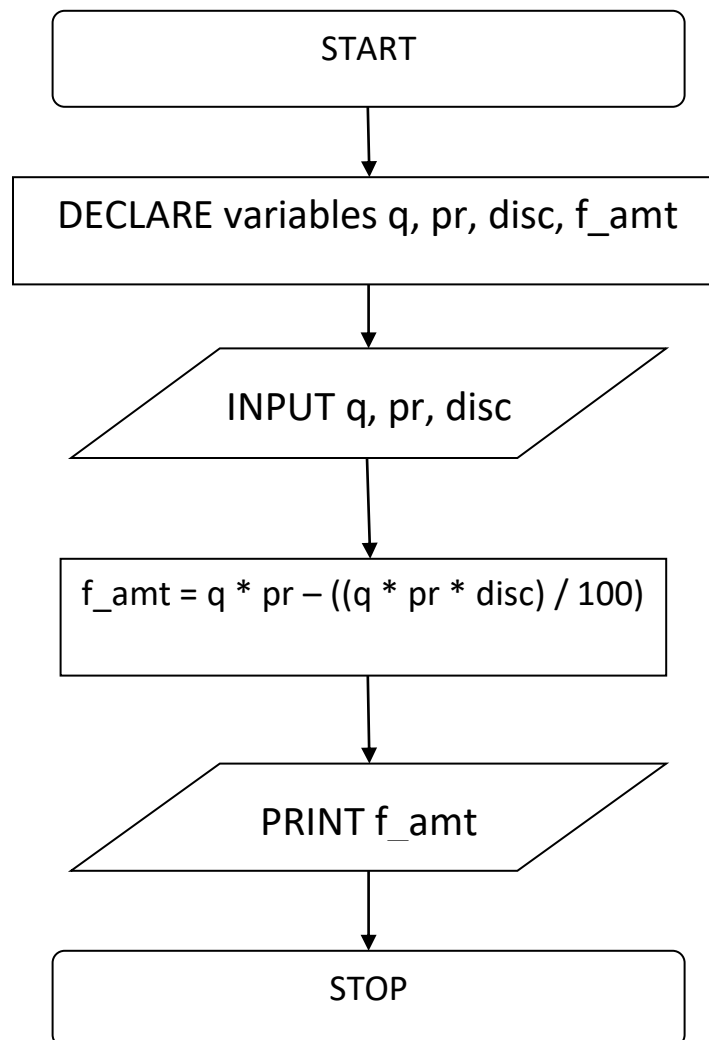
Step 3 : INPUT q, pr, disc

Step 4 : $f_amt = q * pr - ((q * pr * disc) / 100)$

Step 5 : OUTPUT f_amt

Step 6 : STOP

Flowchart :



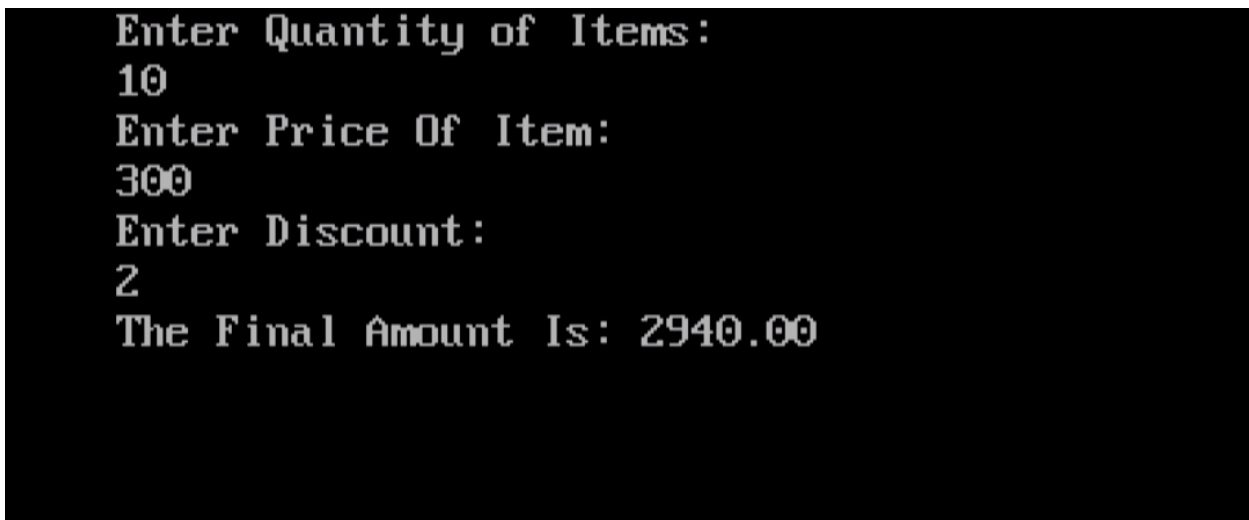
Code :

```
#include<stdio.h>
#include<conio.h>

void main(){

    float Input_quantity,Price,Discount,Final_Amount;
    clrscr();
    printf("Enter Quantity of Items: \n");
    scanf("%f",&Input_quantity);
    printf("Enter Price Of Item: \n");
    scanf("%f",&Price);
    printf("Enter Discount: \n");
    scanf("%f",&Discount);
    Final_Amount = Input_quantity*Price-((Input_quantity*Price*Discount)/100);
    printf("The Final Amount Is: %.2f",Final_Amount);
    getch();
}
```

Output :

A screenshot of a terminal window with a black background and white text. The text shows the execution of a C program. It starts with the prompt 'Enter Quantity of Items:' followed by the input '10'. Then it prompts 'Enter Price Of Item:' followed by the input '300'. Next, it prompts 'Enter Discount:' followed by the input '2'. Finally, it displays the result 'The Final Amount Is: 2940.00'.

```
Enter Quantity of Items:
10
Enter Price Of Item:
300
Enter Discount:
2
The Final Amount Is: 2940.00
```